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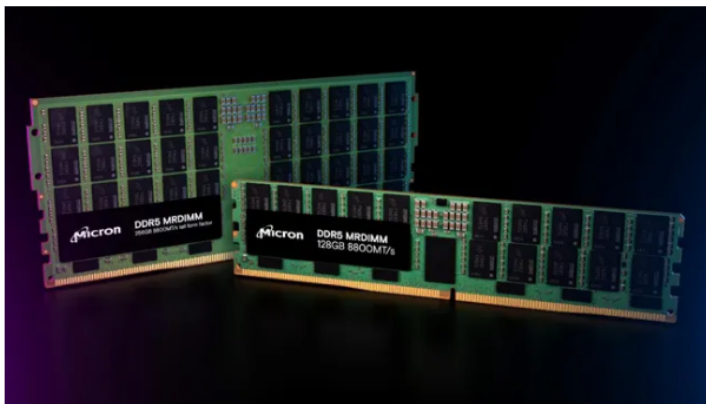
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Micron unveils MRDIMMs for Intel Xeon 6: Up to 256GB DDR5-8800 modules

News

By Anton Shilov | Published July 19, 2024

Next-gen memory for next-gen servers.



(Image credit: Micron)

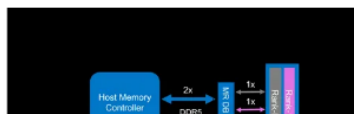


Micron Technology has [introduced](#) its new multiplexed rank dual inline memory modules (MRDIMMs) designed to combine high performance, low latency, high capacity, and predictable power consumption for next-generation [Intel Xeon 6](#) server platforms. As the number of cores per socket rise, the new modules will be particularly useful to ensure decent per-core memory bandwidth, which will be valuable for artificial intelligence (AI) and high-performance computing (HPC) applications.

MRDIMMs are JEDEC-standardized next-generation memory modules that essentially feature two DDR ranks that run in a multiplexed mode to double the speed. To enable such operation, the modules are equipped not only with more memory devices, but also with an MRCD chip that allows simultaneous access to both ranks and MDB chips that enable muxing and demuxing. The host CPU interacts with an MRDIMM module at an 8800 MT/s data transfer rate (for the first generation, the next generation expected to reach 12800 MT/s), but all the components on the module work at half the rate, enabling tighter latencies and keeping power consumption of the module in check.

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What is MRDIMM ?

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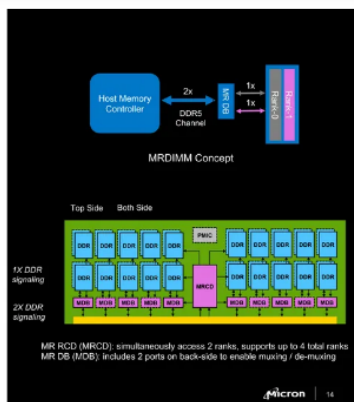
What is MRDIMM ?

[Multiplexed Rank Dual Inline Memory Module](#)

- Data from 2 DDR ranks multiplexed for 2X DDR speed
- Initial speeds up to 8800 MT/s
- Next generation at 12800 MT/s.
- Subsequent generations being aligned in JEDEC



(Image credit: Micron)



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Server memory that delivers performance

Fastest main memory on the planet

- Up to 45% increase in data rate¹
- Up to 39% increase in system bandwidth²

Lowest latency

- Up to 40% lower loaded latency³

Capacity with no limits

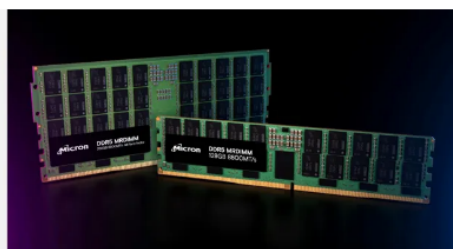
- Up to 39% better performance on 256GB MRDIMM compared to 256GB TSV RDIMMs⁴

Ease of integration

- Standard DDR5 physical and electrical interface
- Broad ecosystem support
- Multi generation scalability without complex expensive packaging like TSV



(Image credit: Micron)



Product leadership delivering highest bandwidth and lowest latency at high capacity leveraging proven DDR5 technology

1. Increase in data rate based on expected future speeds of MRDIMM over RDIMM
2. Bandwidth (data rate) based on expected future speeds of MRDIMM over RDIMM
3. For critical Intel HPC data on opening 128GB MRDIMM (8800MT/s) against 128GB RDIMM (8000MT/s) @ 1TB
4. Read only data based on GenAI platform or different memory stack for MRDIMM (single rank at 8800MT/s) and TSV RDIMM (multiple at 8000MT/s)

Micron | 15

"Micron's latest innovative main memory solution, MRDIMM, delivers the much-needed bandwidth and capacity at lower latency to scale AI inference and HPC applications on next-generation server platforms," said Praveen Vaidyanathan, vice president and general manager of Micron's Compute Products Group. "MRDIMMs significantly lower the amount of energy used per task while offering the same reliability, availability and serviceability capabilities and interface as

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needed bandwidth and capacity at lower latency to scale AI inference and HPC applications on next-generation server platforms," said Praveen Vaidyanathan, vice president and general manager of Micron's Compute Products Group. "MRDIMMs significantly lower the amount of energy used per task while offering the same reliability, availability and serviceability capabilities and interface as RDIMMs, thus providing customers a flexible solution that scales performance. Micron's close industry collaborations ensure seamless integration into existing server infrastructures and smooth transitions to future compute platforms."

Micron Confidential

Micron DDR5 MRDIMM comprehensive portfolio

Delivers the highest performance high-capacity main memory solution for data centers



(Image credit: Micron)

Micron plans to offer a comprehensive line-up of DDR5-8000 and DDR5-8800 MRDIMMs in both standard and tall form factors (TFF) based on 16 Gb, 24 Gb, and 32 Gb DDR5 DRAM devices made using its proven 1β technology. Standard-height MRDIMMs will be available in 32 GB, 64 GB, 96 GB, 128 GB, and 256 GB density without using any 3D-stacked DRAM devices, which means higher performance, lower power, and lower costs. TFF MRDIMMs with a 56.9 mm height and aimed at over 1U servers will be available in 128 GB and 128 GB capacities.

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RAM Benchmark Hierarchy 2026: DDR5, DDR4 for AMD, Intel CPUs



Tall form-factor MRDIMMs have a larger surface area compared to standard-sized modules, which, for the same airflow, results in a 24% lower temperature. This design reduces the likelihood of thermal throttling and enhances energy efficiency, making them particularly suitable for AI and HPC machines.

Upcoming MRDIMMs from Micron and other suppliers (including both first-party makers like Samsung and SK Hynix and third-party suppliers like Adata) are set to be compatible with Intel's Xeon 6 platforms. On the software side of matters, DDR5 MRDIMMs do not require any software development or turning, though expect large server OEMs like Dell, HPE, and Lenovo to run comprehensive tests to ensure compatibility and stable performance across hundreds of workloads.

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Micron's MRDIMMs are currently sampling and are expected to ship in volume in the latter half of 2024. Subsequent generations promise even greater performance, with up to 45% better memory bandwidth per channel – at 12800 MT/s — compared to current RDIMMs at 6400 MT/s.

TOPICS

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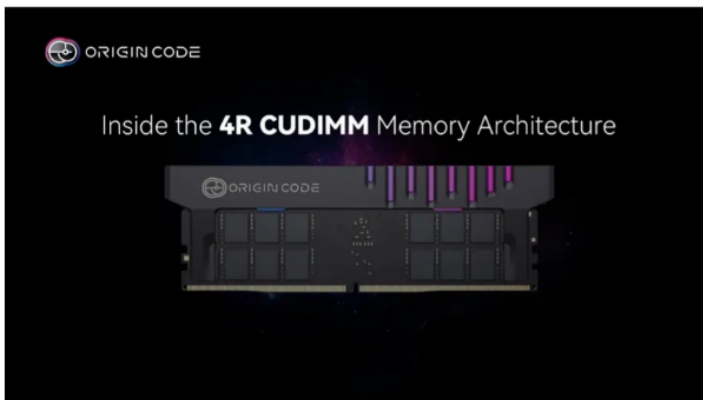


Anton Shilov Contributing Writer



Anton Shilov is a contributing writer at Tom's Hardware. Over the past couple of decades, he has covered everything from CPUs and GPUs to supercomputers and from modern process technologies and latest fab tools to high-tech industry trends.

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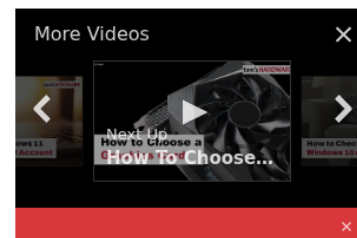


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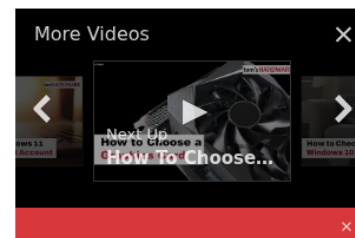
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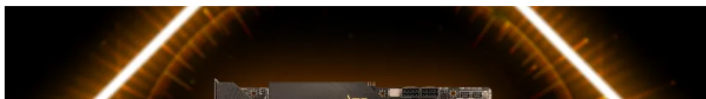


DDR5

Gigabyte announces support for Chinese-made CXMT memory — pushes it to 8200 MT/s on Socket AM5



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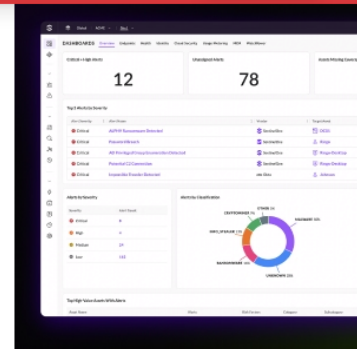
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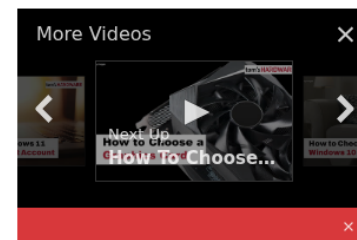


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DDR5

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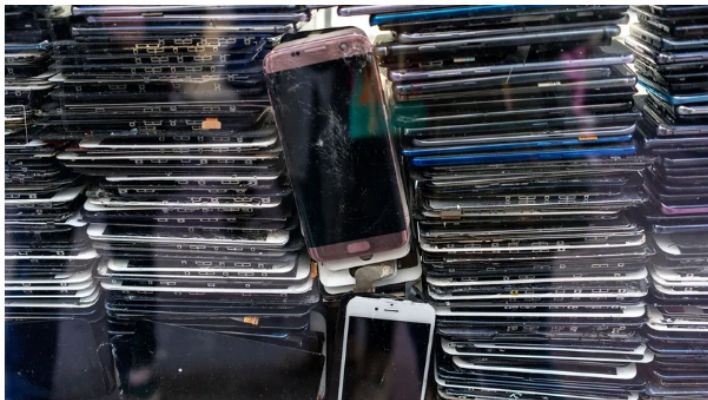


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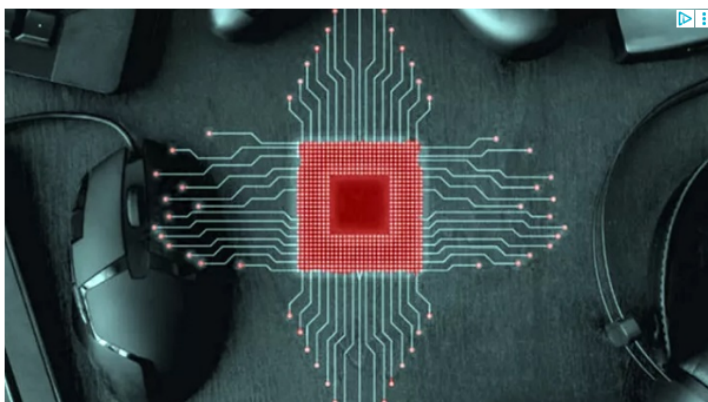


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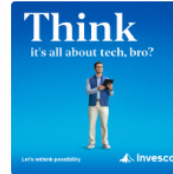
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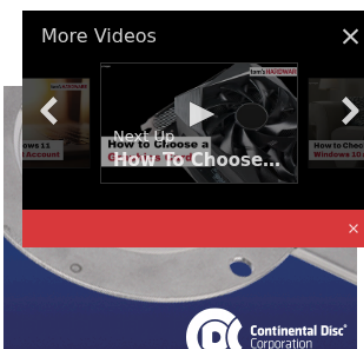
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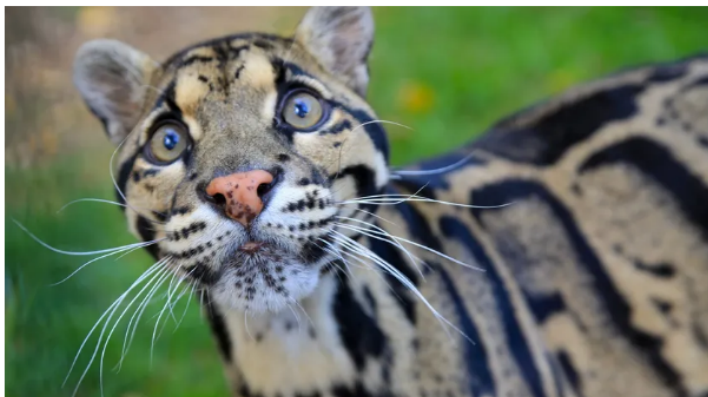
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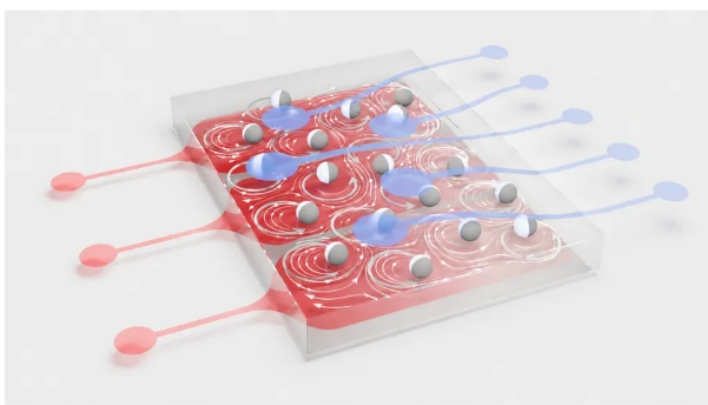


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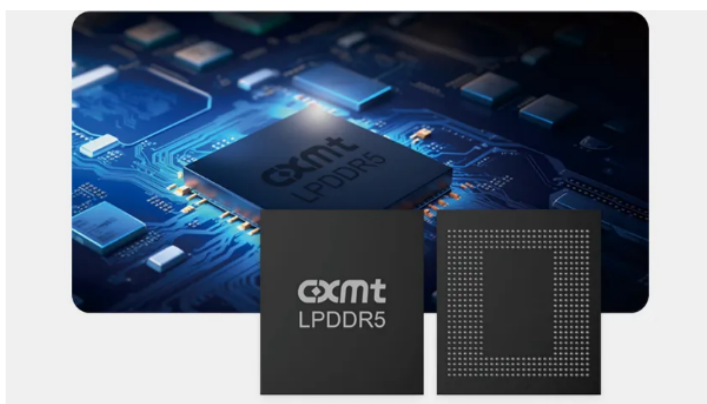
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rluker5

My tuned DDR5 runs at a latency of 55ns. Which isn't great, but seems fairly decent. If these new Mr.DIMMs can knock 40% off of that it would put the latency in L3 cache range.
That sounds too good to be true. But if it is then it would also change how CPUs are built with large L2 taking priority and large L3 becoming irrelevant.

Edit: The Dec 2021 Micron DDR5 DIMMS I have could only do about 65ns latency so 60% of that drops it to 39ns tuned, which is not quite as good as L3 anymore. Hopefully Hynix figures out a way to make these.

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anoldnewb

On

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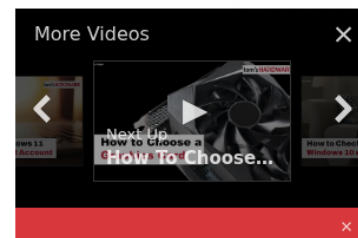
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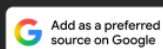
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60% of 55ns is 33 ns. L3 Cache is about 10ns latency (40 cycles at 4 GHz) so there is still a large difference compared to DDR5 memory.
From nice article at chipsandcheese -AMD's Zen 4, Memory Subsystem

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